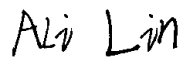


		MODULE TEST REPORT UL 9540A Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems (AACD)	
Project Number. : 4790880512.1 Date of issue..... : 2023-11-10 Total number of pages : 25			
UL Report Office		UL(Changzhou) Quality Technical Service Co., LTD	
Applicant's name		Contemporary Amperex Technology Co., Limited	
Address.....		No.2, Xingang Road,Zhangwan Town, Jiaocheng District Ningde City,352100 Fujian,P.R. China	
Test specification:		4 th Edition, Section 8, November 12, 2019	
Standard..... :		UL 9540A, Test Method for Evaluating Thermal Runaway Fire Propagation in Battery Energy Storage Systems	
Test procedure		8.1 – 8.4	
Non-standard test method		N/A	
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General disclaimer: The test results presented in this report relate only to the sample tested in the test configuration noted on the list of the attachments. UL LLC did not select the sample(s), determine whether the sample(s) was representative of production samples, witness the production of the test sample(s), nor were we provided with information relative to the formulation or identification of component materials used in the test sample(s). The issuance of this report in no way implies Listing, Classification or Recognition by UL and does not authorize the use of UL Listing, Classification or Recognition Marks or any other reference to UL on the product or system. UL LLC authorizes the above named company to reproduce this Report provided it is reproduced in its entirety. UL's name or marks cannot be used in any packaging, advertising, promotion or marketing relating to the data in this Report, without UL's prior written permission. UL LLC, its employees, and its agents shall not be responsible to anyone for the use or non-use of the information contained in this Report, and shall not incur any obligation or liability for damages, including consequential damages, arising out of or in connection with the use of, or inability to use, the information contained in this Report.			

Cell level information		
	Cells in Module:	
	•Manufacturer Name	Contemporary Amperex Technology Co., Limited
	•Part Number	CBDD0
	•Chemistry	Lithium Iron Phosphate
	•Format	Prismatic
	Ratings (Vdc, Ah) :	3.2V, 306Ah
	Was the cell certified? :	Yes
	Standard the cell was certified to:	UL1973
	Organization that certified the cell:	MH62898
	Average cell surface temperature at gas venting, °C:	154
	Average cell surface temperature at thermal runaway, °C:	241
	Gas Volume:	204L
	Lower flammability limit (LFL), % volume in air at the ambient temperature:	8.595
	Lower flammability limit (LFL), % volume in air at the venting temperature:	7.24
	Burning velocity (S _u) cm/s:	54.20
	Maximum pressure (P _{max}) psig:	102.74
Cell Gas Composition:		
Gas		Measured %
Carbon Monoxide	CO	14.596
Carbon Dioxide	CO ₂	26.925
Hydrogen	H ₂	43.066
Methane	CH ₄	7.051
Acetylene	C ₂ H ₂	0.119
Ethylene	C ₂ H ₄	3.289
Ethane	C ₂ H ₆	1.060
Propylene	C ₃ H ₆	0.686
Propane	C ₃ H ₈	0.260
-	C ₄ (Total)	0.865
-	C ₅ (Total)	0.399
-	C ₆ (Total)	0.148
1-Heptene	C ₇ H ₁₄	0.025
Styrene	C ₈ H ₈	0.013
Benzene	C ₆ H ₆	0.082
Toluene	C ₇ H ₈	0.012
Dimethyl Carbonate	C ₃ H ₆ O ₃	1.304
Ethyl Methyl Carbonate	C ₄ H ₈ O ₃	0.101
Total	-	100.000

Module level Information	
Model No:	M1306P05L01
Ratings (Vdc, Ah) :	166.4V,306Ah
Module cell configuration (xS/yP)	52S 1P
Module dimensions (W x D x H (mm)) :	810*1156*241.5
Module weight (kgs) :	330
Module enclosure material:	Bottom enclosure: Al3003 Top Plastic enclosure: PP-B-S01
Was the module certified? :	No
Standard the module was certified to:	N/A
Organization that certified test item:	N/A
Cell failure test method performed for the module level (summary of method and test clause): ☒ External heating using thin film with 4°C to 7°C thermal ramp. ☐ Nail Penetration ☐ Overcharge ☐ External short circuit (<i>X Ω external resistance</i>) ☐ Others	
Description of method used to fail cells if other than external thin film heater with thermal ramp :N/A	
Description of components employed within the module that serve to suppress propagation (fire protection features). N/A	
Number of initiating cells failed to achieve propagation.	1
Thermal Runaway Propagation:	Initiating cell went into thermal runaway and propagated to two adjacent cells
Maximum Smoke Release Rate (m²/s)	1.81
Total Smoke Released: (m²)	174.72
Total smoke released duration	0:41:24~02:00:00
Peak Chemical Heat Release Rate: (kW):	No flaming occurred
External Flaming:	No external flaming occurred
Location(s) of Flame Venting:	No flaming occurred
Flying Debris:	No flying debris occurred
Re-ignitions:	No further re-ignitions were observed during post test observation.

Summary of Module level test Gas Analysis Data:				
Gas Analysis: ☒ Flame ionization detection ☐ Fourier-Transform infrared Spectrometer ☒ Hydrogen Sensor (palladium-nickel, thin-film solid state sensor) ☒ White light source with photo detector (smoke release rate)				
• Gas Composition & Volume for Each Compound (Pre-flaming and After flame):				
Gas Compound	Gas Type	Pre-Flaming (L)	Flaming (L)	Minimum detectable flow rate (LPM)
Total Hydrocarbons (Propane Equivalent)	Hydrocarbons	261.95	No flaming	0.51
Carbon Dioxide	Carbon Containing	246.40	No flaming	1.24
Carbon Monoxide	Carbon Containing	78.75	No flaming	0.60
Hydrogen	Hydrogen	263.97	No flaming	13.43
Summary of Module testing:				
Performance Criteria in accordance with Clause 8.4 and Figure 1.1: ☒ The effects of thermal runaway was contained by the module design; ☐ Cell vent gas (based upon the cell level test) was non-flammable				
Necessity of a unit level test				
☒ The performance criteria of the module level test as indicated in 8.4 and as shown in Figure 1.1 of UL 9540A 4th edition has not been met, therefore unit level testing in accordance with UL 9540A will need to be conducted on a complete unit employing this module. ☐ The performance criteria of the module level test as indicated in 8.4 and as shown in Figure 1.1 of UL 9540A 4th edition has been met, therefore unit level testing in accordance with UL 9540A need not be conducted.				
Testing Laboratory information				
Testing Laboratory and testing location(s):				
Testing Laboratory:		Beijing Building Materials Testing Academy		
Testing location/ address		Block 1, B15 Yaxin Road, Doudian Town, Fangshan district, Beijing 102402, CN		
Tested by (name, signature)		Zhang Qi, Ali Lin		
Witnessed by (for 3 rd Party Lab Test Location) (name, signature)		N/A	N/A	
Project Handler (name, signature).....		Ali Lin(T)/Gavin Chen	 	

Reviewer (name, signature)	Albert He	
List of Attachments (including a total number of pages in each attachment):		
Attachment A: Module Conditioning (Charge/discharge) Profiles - (<i>Pages 16 through 17</i>) Attachment B: Module Construction Photos - (<i>Pages 18 through 18</i>) Attachment C: Module Instrumentation Photos - (<i>Pages 19 through 20</i>) Attachment D: Module and Initiating Cell(s) Temperature Profiles During Testing - (<i>Pages 21 through 21</i>) Attachment E: Module Testing Photos - (<i>Pages 22 through 23</i>) Attachment F: Module Gas Flow Rate and Heat Release Profiles - (<i>Pages 24 through 25</i>)		

Photo(s) of module:**Test Item Charge/Discharge Specifications:**

• Charge power, kW:	25.460
• Charge current, A:	153
• Standard Full charge Voltage, Vdc:	189.8
• Charge temperature range, °C:	0~60
• End of charge Voltage, Vdc	Any cell reaches of 3.65 or module voltage 189.8
• Discharge power, kW:	25.460
• End of discharge voltage, Vdc:	Any cell reaches of 2.5 or module voltage 130
• Discharge temperature range, °C:	-20~60

Test item particulars..... :	
Possible test case verdicts: - test case does not apply to the test object: N/A - test object does meet the requirement.....: P (Pass) - test object does not meet the requirement: F (Fail) - test object was completed per the requirement...: C(Complete) - test object was completed with modification.....: M(Modification) Testing.....: N/A Date of receipt of test item: 2023-06-01 Date (s) of performance of tests: 2023-08-12	
General remarks:	
"(See Enclosure #)" refers to additional information appended to the report. "(See appended table)" refers to a table appended to the report. Throughout this report a point is used as the decimal separator.	
Manufacturer's Declaration of samples submitted for test:	
The applicant for this report includes samples from more than one factory location and a declaration from the Manufacturer stating that the sample(s) submitted for evaluation is (are) representative of the products from each factory has been provided:	☐ Yes ☒ Not applicable
Name and address of factory (ies) Contemporary Amperex Technology Co., Limited Factory address1: No.2 Xingang Road Zhangwan Town, Jiaocheng District, Ningde, 352100 Ningde, Fujian, PEOPLE'S REPUBLIC OF CHINA	
General product information and other remarks:	
Battery module model M1306P05L01 employs cell Models CBDD0 manufactured by Contemporary Amperex Technology Co Limited, rated 166.4V, 306Ah.	

5.0	CONSTRUCTION		Verdict
5.2	Module Construction		—
5.2.1, 5.2.3	Construction information	See Test Item Description at the beginning of this report	—
	General layout of module contents	See Attachment B	—
5.2.2	Module certified to UL 1973	No	
	Organization that certified module:	N/A	—
6.0	PERFORMANCE		Verdict
6.1	General		
8.1	Samples		
8.1.1	Samples conditioned through charge discharge cycling a minimum of 2 cycles.	See Attachment A for profiles See Table 1 for specifications See also Table 2 The module voltage was checked before the test, and the voltage did not drop further compared to 1h to 8h after cycles, which was judged acceptable.	M
8.1.2	100% SOC and stabilize from 1h to 8 h before testing		
8.1.3	Electronic controls such as BMS not relied upon during testing.		C
8.2	Test Method		
8.2.1	Ambient indoor laboratory conditions: 25 ±5°C (77 ±9°F) ≤50 ±25% RH at the initiation of the test.	See Table 3	C
8.2.2	Test conducted under a smoke collection hood appropriately sized for the module		C
8.2.3	The weight of the module was recorded before and after testing, (kg)	See Table 11	C
8.2.4	A sufficient number of cells were forced into thermal runaway to create a condition of cell to cell propagation within the module.	See Attachment C and F See Tables 4 and 5	C
	The location of the cell(s) forced into thermal runaway were selected to present the greatest thermal exposure to adjacent cells	See Attachment C for figures showing location within the module of the cell(s) forced into thermal runaway	C
8.2.5	The method used to initiating thermal runaway in the cell(s) were in accordance with 7.2	See Summary of Cell Testing at the beginning of this report.	C
8.2.6	The occurrence of thermal runaway was verified	See Test Results from Cell Level Test from the beginning of this report See Attachments D and F	C
8.2.7	The module was placed on top of a non-combustible horizontal surface with the module orientation representative of its intended final installation.	See Attachment E	C

8.2.8	The chemical heat release rate of the module was measured with oxygen consumption calorimetry	See Table 10 See Attachment F	C
8.2.9	The chemical heat relate rate was measured for the duration of the test	See Attachment F	C
8.2.10	The chemical heat release rate was measured using the following equipment: • Paramagnetic oxygen analyser • Non-dispersive infrared carbon dioxide and carbon monoxide analyser • Velocity probe • Type K thermocouple	See Attachment F	C
	The instrumentation was located in the exhaust duct of the heat release rate calorimeter at a location that minimizes the influences of bends or exhaust devices.		C
8.2.11	The chemical heat release rate at each of the flows was calculated in accordance with 8.2.11.	See Attachment F	C
8.2.12	The hydrocarbon content of the vent gas was measured using flame ionization detection.	See Table 8 and 9	C
	Hydrogen gas shall be measured with a palladium-nickel thin-film solid state sensor.	See Table 9	C
8.2.13	The hydrocarbon content of the vent gas may also be measured using a Fourier-Transform Infrared Spectrometer with a minimum resolution of 1 cm ⁻¹ and a path length of at least 2 m (6.6 ft), or equivalent gas analyzer.	FTIR analysis was not used in accordance with the Certification Requirement Decision: Corrections to gas measurement methods to make FTIR as an option for measuring hydrocarbon contents of gas emissions and to include Hydrogen measurements during the Unit Level Test. FTIR was considered redundant to the other gas measurement methods used.	N/A
	Vent gas velocity and temperature measurements respectively were obtained in the exhaust duct of the heat release rate calorimeter using equipment specified in 8.2.10.		C
8.2.14	The light transmission in the exhaust duct of the heat release rate calorimeter was measured using a white light source and photo detector for the duration of the test.		C
8.2.15	Smoke release rate was calculated as outlined in 8.2.15	See Table 10 See Attachment F	C
8.3	Module level test report		
	a. Module manufacturer and model number; b. Number of cells in module; c. Module configuration;	See Test Item Description in beginning of this report.	C

	d. Module construction features;	See Attachment C See Critical Components Table	C
	e. Module voltage corresponding to the tested SOC;	See Table 3 See Attachment F	C
	f. Thermal runaway initiation method used;	See Attachment C and F	C
	g. Heat release rate versus time data;	See Table 10 See Attachment F	C
	h. Flammable gas generation and composition data;	See Attachment F See Tables 8 and 9	C
	i. Peak smoke release rate and total smoke release data.	See Table 10	C
	j. Observation(s) of flying debris or explosive discharge of gases;	See Table 12	C
	k. Observation(s) of sparks, electrical arcs, or other electrical events;	See Table 12	C
	l. Identification/location of cells(s) that exhibited thermal runaway within the module;	See Tables 4 and 5	C
	m. Locations and visual estimations of flame extension and duration from the module;	See Attachments E and F See Table 7	C
	n. Module weight loss;	See Table 11	C
	o. Video of the test.		C
8.4	Performance – Module level		
8.4.1	The following performance conditions are met during the module level test: a) Thermal runaway is contained by module design;	Thermal runaway is contained by module design.	P
	b) Cell vent gas is nonflammable as determined by the cell level test	Cell gas report show the cell gases are flammable. See Cell Gas Composition Table.	F

Table 1 – Specified conditioning parameters

Charging:		Discharging:	
Current (CP), kW	See Attachment A	Current (CP), kW	25.460
Standard full Charge Voltage, Vdc	Any cell reaches of 3.65V or total voltage reach 189.8V	End of discharge voltage, Vdc	Any cell reaches of 2.5V or total voltage reach 130V
Cutoff Voltage, V	Any cell reaches of 3.65V or total voltage reach 189.8V	Cutoff Voltage, V	Any cell reaches of 2.5V or total voltage reach 130V
Charging Test Ambient, °C	0~60	Discharging Test Ambient, °C	-20~60
Refer to Attachment A for charge/discharge profiles for the module.			

Table 2 – Charge completion and module test initiation times

Charge Completion Date and Time	Module Test Date and Time
2023-08-11 12:06:22 AM	2023-08-12 12:38:43 AM

Table 3 - Test Initiation Details

Module No.: M1306P05L01	
Test Date	2023-08-12
Test Start Time	12:38:43 PM
Initial Lab Temperature	26.8°C
Initial Relative Humidity	71%
Module OCV at Start of Test, Vdc	173.8

Table 4 – Approximate time of thermal runaway propagation through module

Time to thermal runaway	Location
0:42:20	Cell 20
0:43:40	Cell 21
0:47:34	Cell 19

Table 5 – Test overview timeline

Time (HH:MM:SS)	Event	Description
00:00:00	Test Start	The test was started and the heater was turned on to heat the initiating cell (Cell 20) at a ratio of 4 ~ 7 °C/min. See Figure (a)
00:41:24	Venting of initiating Cell	Initiating cell (Cell 20) vented at around 157.3°C measured through T2-2 by an indication of sudden dip in cell's temperature curve. See Figure (b)
00:42:20	Thermal runaway of initiating cell	Initiating cell (Cell 20) vented at around 181.2°C measured through T2-2 by an indication of sudden dip in cell's temperature curve. See Figure (c)
00:43:40	Thermal runaway of adjacent cell	Thermal runaway propagated to nearby cell (cell 21). Smoke is released from the vent accompanied by electrolyte and molten material outflow. See Figure (d)
00:47:34	Thermal runaway of adjacent cell	Thermal runaway propagated to nearby cell (cell 19). Smoke is released from the vent accompanied by electrolyte and molten material outflow. See Figure (e)
02:00:00	Test Termination	Data acquisition was stopped. The module was left in the test overnight and with video monitored. See Figure (f)

Table 6 – Gases measured and measurement methods used in unit level testing			
Measurement Method	Gases Measured	Chemical Formula	Gas Type
Flame Ionization Detection (FID)	Total Hydrocarbons	-	Hydrocarbons
Solid-state Hydrogen Sensor	Hydrogen	H ₂	-
Non-dispersive infrared spectroscopy (NDIR)	Carbon Dioxide	CO ₂	Carbon Containing
	Carbon Monoxide	CO	Carbon Containing
# - This table was modified to reflect the gases measured during testing.			

Table 7 - Gas generation periods	
Time	Condition
0:41:24 to 02:00:00	Pre-Flaming
N/A	Flaming
External Flaming of Gas	
Condition	Duration (hh:mm:ss)
External Flaming of Vent Gases:	N/A

Table 8– Summary of battery gas volumes for deflagration hazard calculations				
Gas Component	Gas Type	During Pre-flaming (L)	During Flaming (L)	Minimum detectable flow rate(LPM)
See Table 9				

Table 9– Summary of battery gas volumes for deflagration hazard calculations				
Gas Component	Gas Type	During Pre-flaming (L)	During Flaming (L)	Minimum detectable flow rate (LPM)
Total Hydrocarbons (Propane Equivalent)	Hydrocarbons	261.95	No flaming	0.51
Carbon Dioxide	Carbon Containing	246.40	No flaming	1.24
Carbon Monoxide	Carbon Containing	78.75	No flaming	0.60
Hydrogen	Hydrogen	263.97	No flaming	13.43

Table 10 – Smoke and heat release rate			
Heat Release Rate (HRR)		Smoke Release Rate (SRR)	
Peak Chemical HRR (kW)	0 (No flaming)	Maximum SRR (m ² /s)	1.81
		Total Smoke Released (m ²)	174.72
Note: The SRR sensor was not wiped clean before the test, leading to an abnormal drift of the base value during the test. The SRR data is for reference only, not affect the module level performance Criteria.			

Table 11 – Module Weight During Test, kg

Before Test:	328.5kg
After Test:	326.5kg
Weight Loss:	2kg

Table 12 – Other Observations during module test

	Observed, Yes/No	Location
Flying debris	No	N/A
Explosive discharge of gas	No	N/A
Sparks or electrical arcs	No	N/A

TABLE: Critical components information					
Object / Part No.	Manufacturer/ trademark	Type / model	Technical data	Standard	Mark(s) of conformity
Cells	Contemporary Amperex Technology Co., Limited	CBDD0	3.2V, 306Ah	UL1973	UL MH62898
Module Model	Contemporary Amperex Technology Co., Limited	M01306P05L01	Nominal voltage: 166.4V Rated capacity: 306Ah	-	-
Metal Enclosure	LONGGUANG ELECTRONICS GROUP CO.,LIMITEDFUJIAN YUANSHENG AUTO PARTS TECHNOLOGY CO., LTD. FuJian Junpeng Communication Technology Co.,Ltd	-	Material: Al3003 Thickness: 1.4mm/1.0mm Size: 1145*810*59.5mm	-	-
Plastic enclosure	Hunan Jiyongsheng Co., Limited	PP	Fire rating: V-0 Maximum ambient temperature: 85°C Min THK: 1.1mm	UL 94	Intertek 20890002-MT-3
External Connctors	Sichuan Recodeal Interconnect System Co., Ltd	REA4-WTa-95-A	Voltage: 1500Vdc Current: 350A ambient temperature: -40~85 °C	EN 61984:2009	TUV J 50586193
PCB material of CSC	VICTORY GIANT TECHNOLOGY (HUIZHOU) CO LTD	SH10	V-0, 130 °C, CTI:0/600	UL 94	UL E248779
Internal connecting wires for HV	DONGGUAN NISTAR TRANSMITTING TECHNOLOGY CO INC	UL3932	2000V d.c., 125°C, 95mm2	UL 758	UL E214184
Plastic material for Cover of CSC	Pulse Electronics Corporation	HM1317HL	Class B	-	-

Voltage sampling wire	DONG GUAN NISTAR TRANSMITTING TECHNOLOGY CO.,INC.	3666	32awg, XLPE, 105°C, 600 Vac.	UL 758	UL E214184
Temperature sampling wire	TAIZHOU RISHUN ELECTRIC APPLIANCE DEVELOPMENT CO LTD	4484	32 - 16 AWG 150°C ,300 Vac.	UL 758	UL E237675
NTC Sensor	SEMITEC CORP	103AP-2	10kΩ±0.5%/1%, -50°C to 110°C	UL 1434	UL E92669
Fuse for one module	XI'AN SINFUSE ELECTRIC CO LTD	RSZ307-1-TMZ-N	200 – 800A, 250 Vdc.	UL 248-1	UL E353337
Plastic cover for fuse	Ningbo Yosun Auto Parts Co.,Ltd	PA66-640G00	V-0, 130°C, thickness: 2mm	UL 94 UL 746B	UL E225348
Explosion relief valve	0000013358	PDA3K 50G -	Discharge pressure: 3kPa Plastic material: PC+GF10/PC Aluminum alloy material : AL 6063 Sealing ring material: silicon rubber Membrane material: e-PTFE	---	---

Attachment A: Module Conditioning (Charge/discharge) Profiles - (Pages 16 through 17)

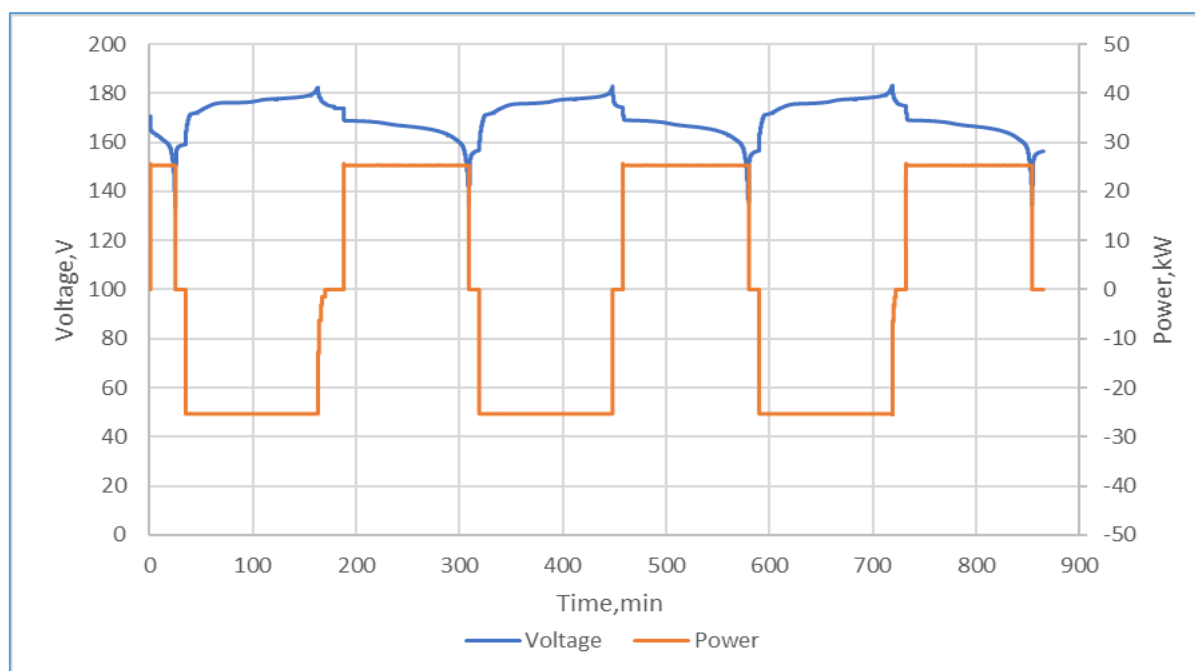
Charge/Discharge parameters:

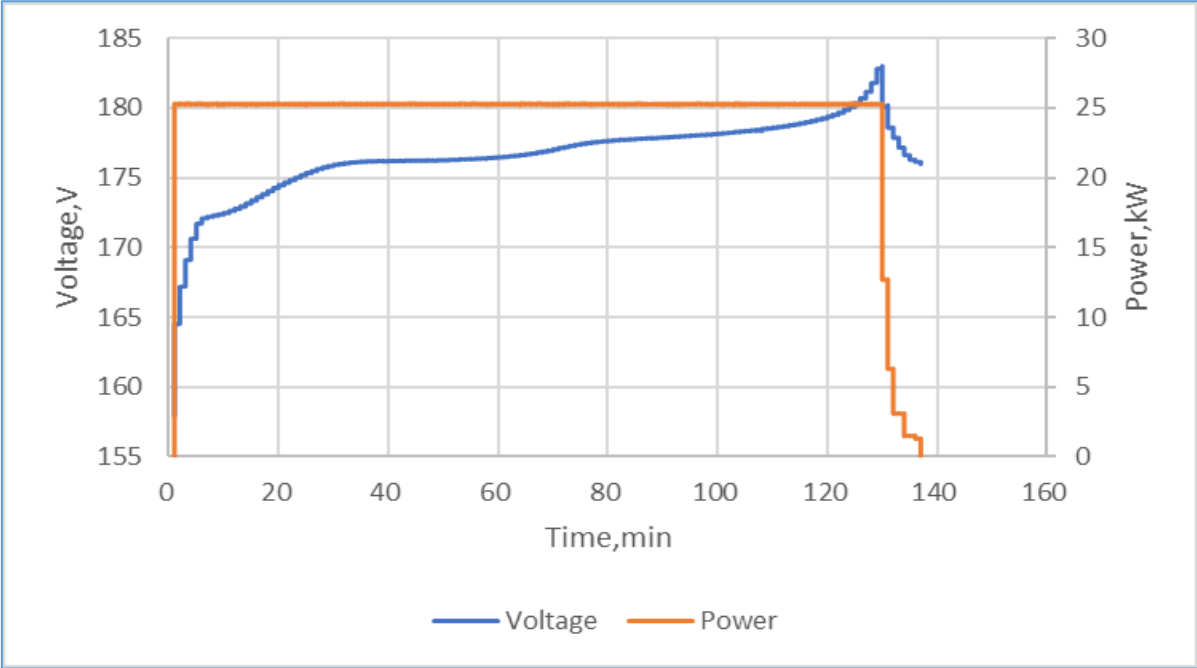
BMS power reduction charging strategy is as follows:

Initially charge at 0.5P(25.46kW) until the battery voltage is 3.65 V, then reduce the charging power to 0.25 P; while any cell voltage is 3.65 V, reduce the charging power to 0.125 P; while any cell voltage is 3.65 V then charge at 0.0625 P and 0.03125 P to 3.65 V in turn.

Then rest for 10min, and discharge at 0.5 P power until the voltage is 2.5 V.

Module charge to 100% SOC: 0.5P(25.46kW) constant power charge until any cell reaches of 3.65V or total voltage reach 189.8V.





Attachment B: Module Construction Photos - (Pages 18 through 18)

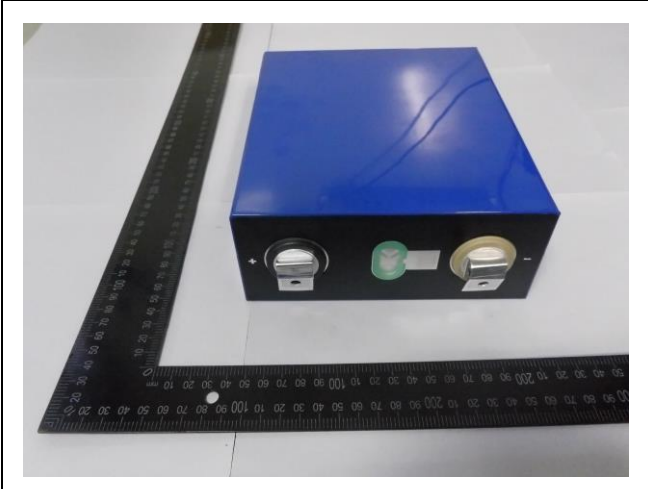


Figure 1 - Photo of cell – Wide side view

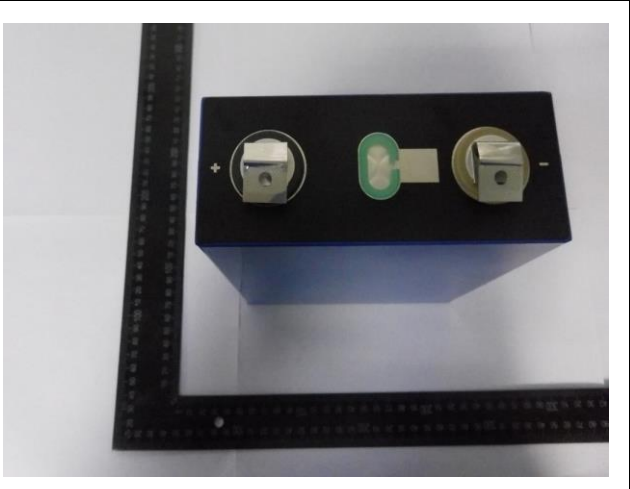


Figure 2 - Photo of cell – Top view

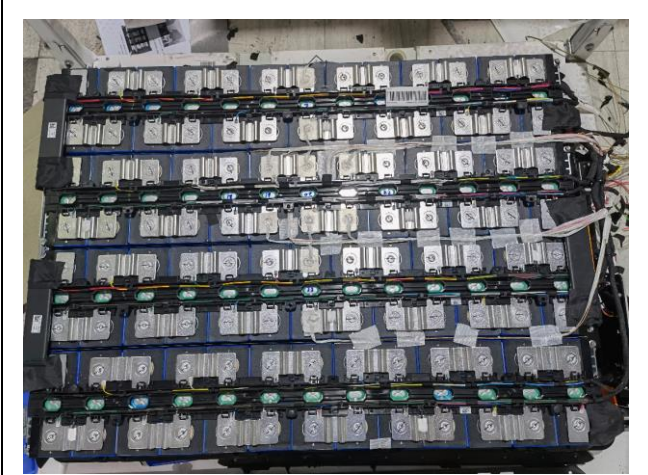


Figure 3 - Photo of module – Internal view



Figure 4 - Photo of module – Front view



Figure 5 – Photo of module – Rear view



Figure 6 - Photo of module – Side view

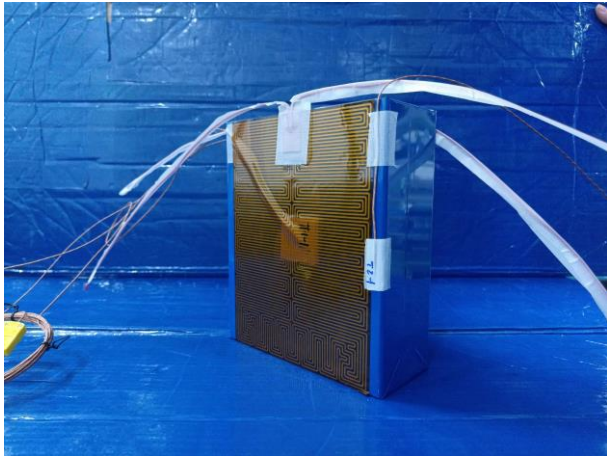
Attachment C: Module Instrumentation Photos - (Pages 19 through 20)

Figure 7 – Cell with heaters. Two pieces of 152.4 mm by 203.2 mm heaters for each sample.



Figure 8 – Module with heater, TCs and Voltage Sampling wires.

Note: The thermocouple T1-1 was used to control the supply power to the heater to keep the heating rate at 4 ~ 7 °C/min. T2-1 and T2-2 were used to represent the temperature of initiating cell.

T1-1, on the wide side surface center of cell 20, between the cell and heater.

T1-2 on the other wide side surface center of cell 20, between the cell and heater.

T2-1, T2-2, on each wide side surface center of cell 20, not covered by heater.

V1, V2, V3, V4, V5 was the voltage of cell 20, 19, 21, 7, 33 separately.

T3-1, T3-2, T4-1, T4-2, T5, T6, T7-1, T7-2, T8-1, T8-2 were attached on the wide side surface center of cells shown in below illustration.

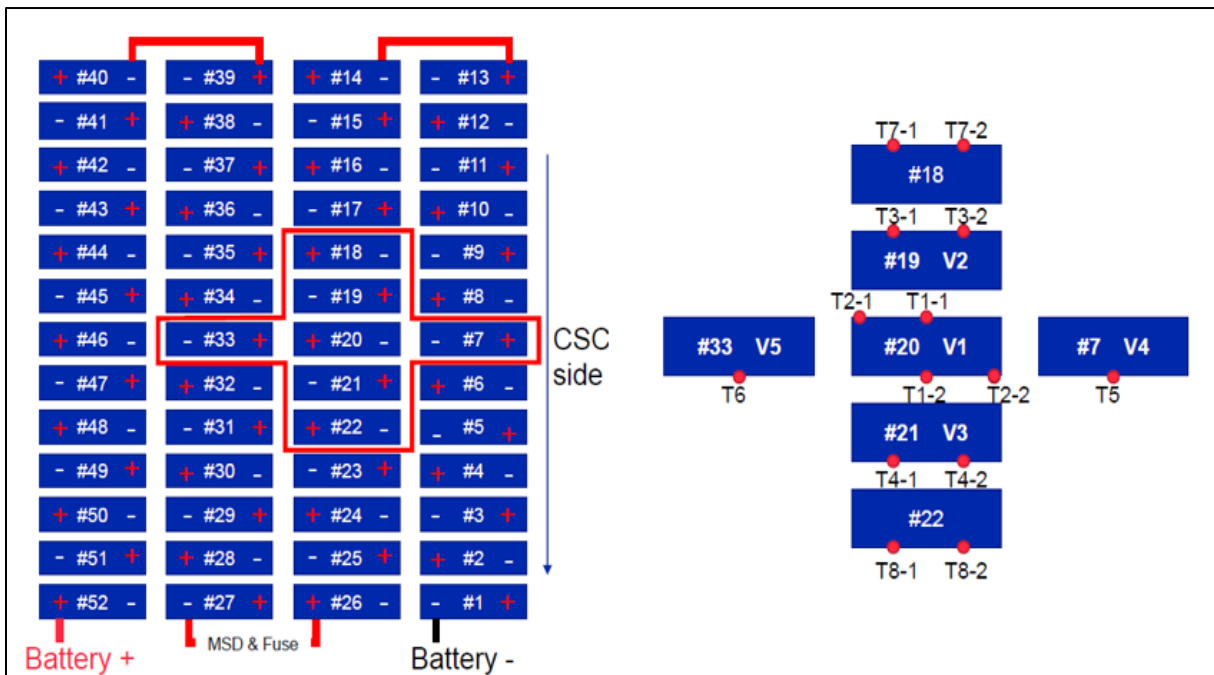


Figure 9 – Locations of TCs



Figure 10 – Module on the test platform – Side view 1



Figure 11 – Module on the test platform – Side view 2

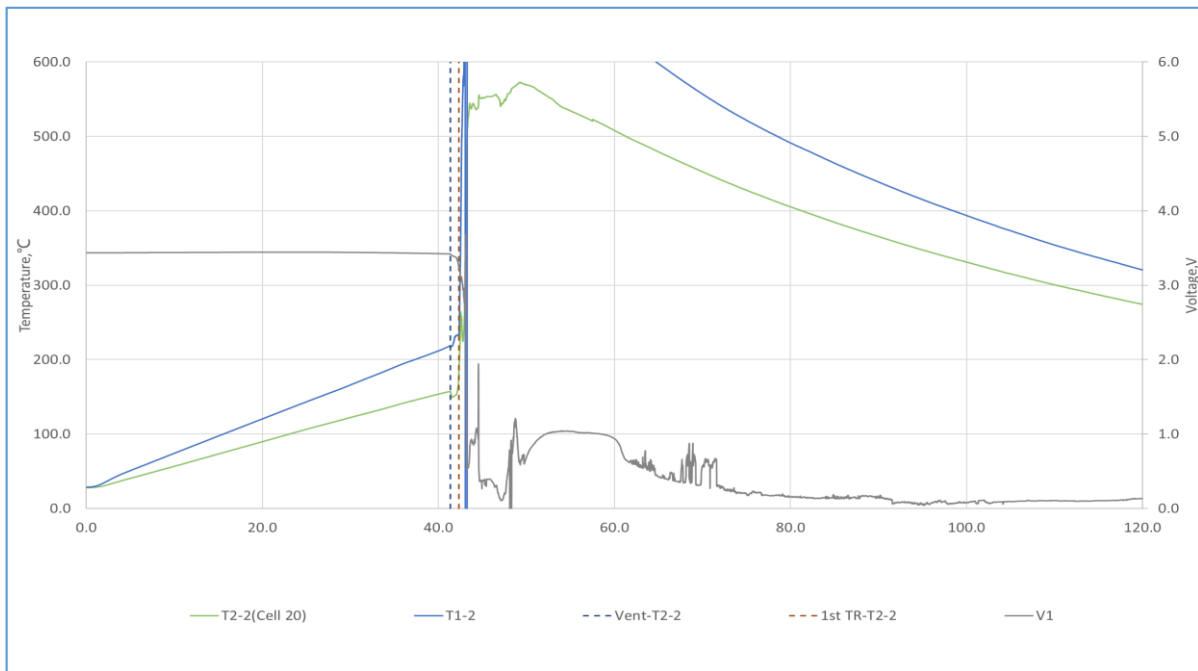
Attachment D: Module and Initiating Cell(s) Temperature Profiles During Testing - (Pages 21 through 21)

Figure 12: Initiating cell Temperature and Voltage Profiles During Testing

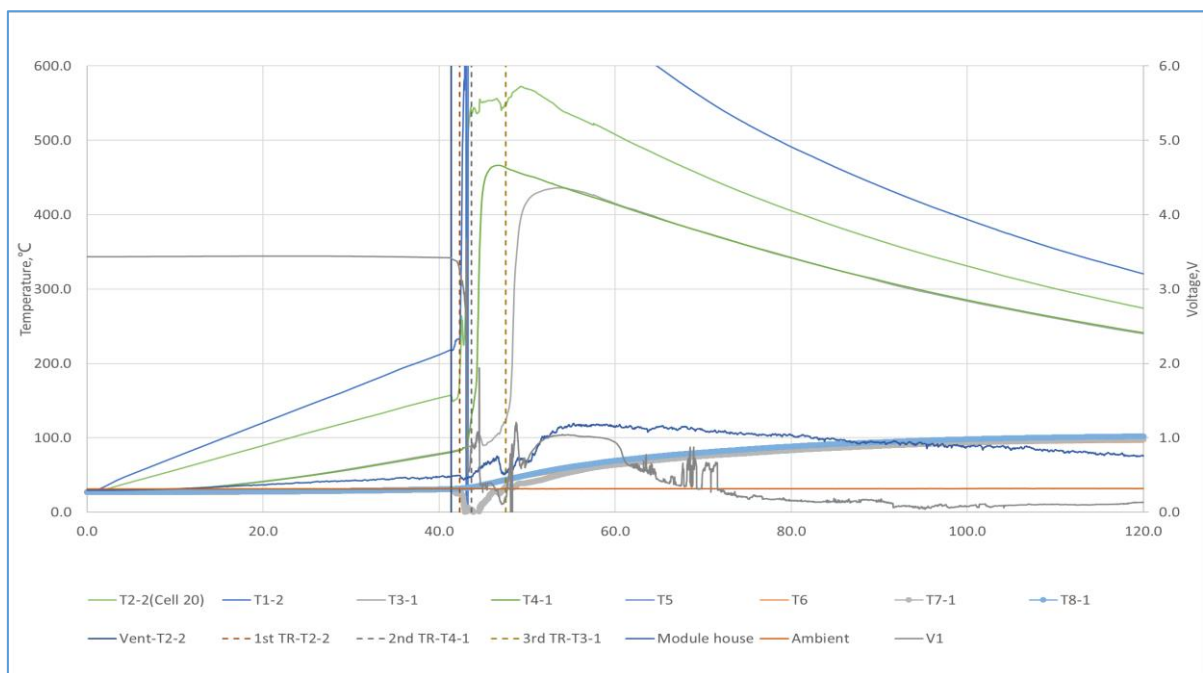
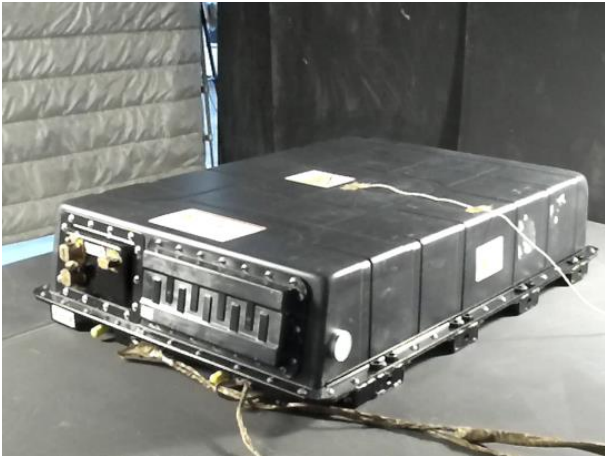


Figure 13: Temperature Profiles Describing Cell to Cell Propagation

Attachment E: Module Testing Photos - (Pages 22 through 23)

(a) Test start
[00:00:00]



(b) Initiating cell 20 Venting
[00:41:24]



(c) Thermal runaway of initiating Cell 20
[00:42:20]



(d) Thermal runaway of adjacent Cell 21
[00:43:40]

	
(e) Thermal runaway of adjacent Cell 19 [00:47:34]	(f) Test Termination [02:00:00]

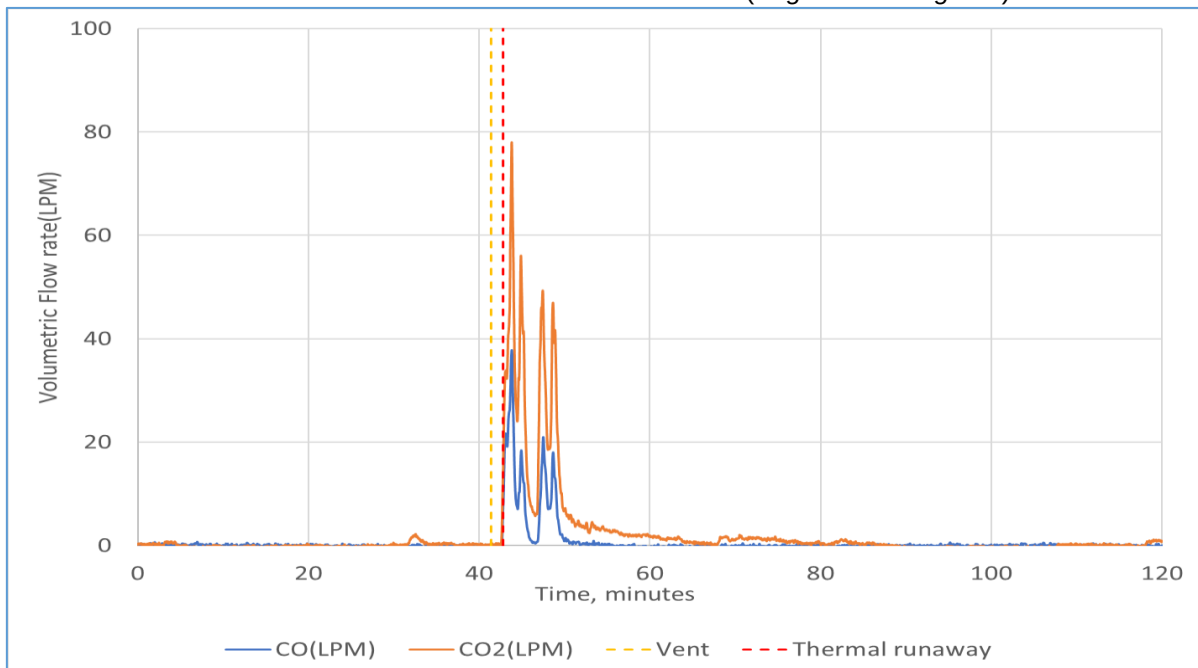
Attachment F: Module Gas Flow Rate and Heat Release Profiles - (Pages 24 through 25)

Figure 14 - CO, CO2 Volumetric flow rates

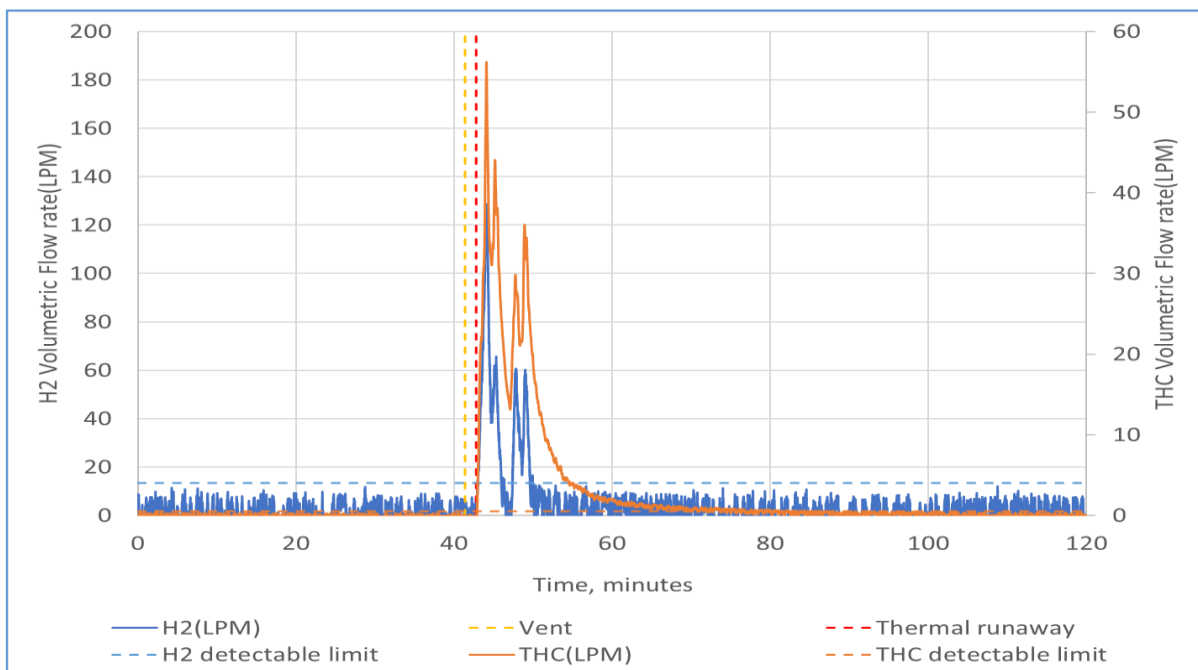


Figure 15 - THC, H2 Volumetric flow rates

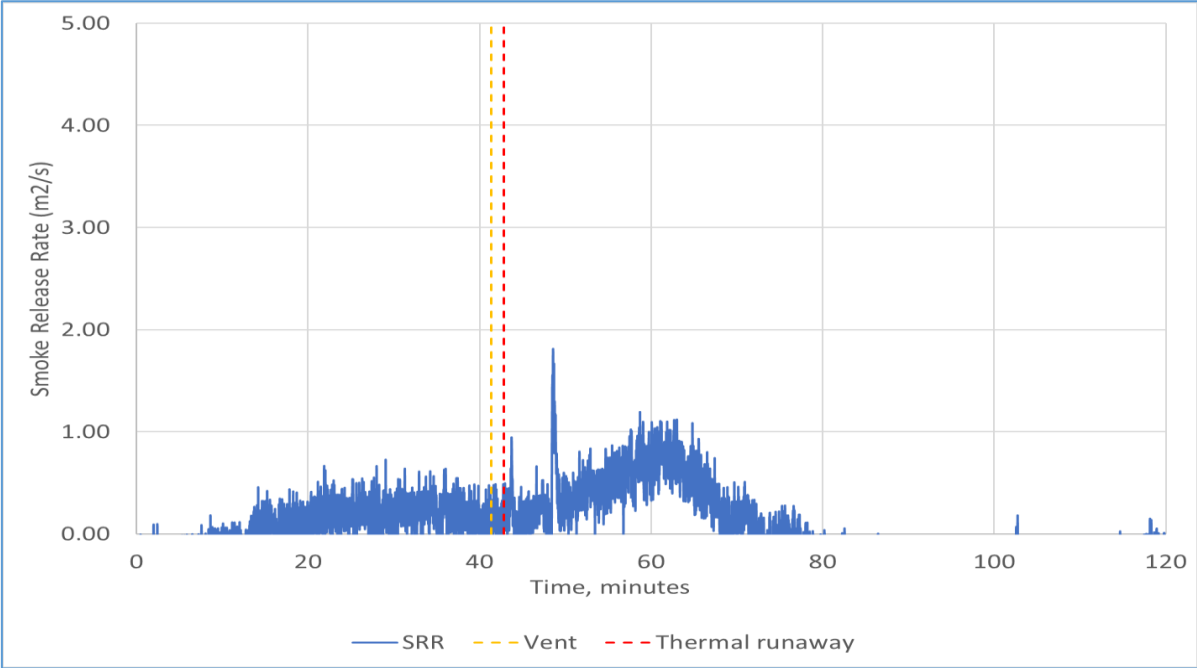


Figure 16 – Smoke release rate